19 August

Making of a scientist

I Answer the following Reference to Context:-

1 "Then ---------- something."

a Who is the speaker of these lines?
(i) Richard Ebright
(ii) Miss. Ebright
(iii) James R. Wong
(iv) Dr. Egbert

b What was the speaker's emotional state?
Ans The speaker felt inspired and deeply impressed by Ebright's curiosity, determination, and passion for science, admiring his journey from a curious child to a dedicated scientist.

c Unlike everybody the speaker
(i) Did not win anything
(ii) could not participate
(iii) was not that good
(iv) all of the above

d When did he realise what real science is?
Ans He realised what real science is when he began conducting experiments, not just collecting butterflies - especially after working on the viral DNA project with Dr. Urquhart's guidance.

II Answer the following Short Answers:-

Q1 What role did Ebright's mother play in his success?
Ans Ebright's mother nurtured his curiosity by providing books, equipment, and encouragement. She took him on trips, bought him a microscope, introduced him to scientific ideas, and kept him engaged with challenging projects. Her constant support and guidance helped shape his discipline, creativity, and passion for science, laying the foundation for his later achievements.

Q2 What are the essential qualities for becoming a scientist according to Ebright's teacher?
ans According to Ebright's teacher, the essential qualities for becoming a scientist are curiosity, an eagerness to learn, and the drive to seek answers. A scientist must not merely collect facts but try to understand their meaning, think creatively, and ask "Why?" and "How?" This combination of inquisitiveness, and critical thinking is what leads to true scientific discovery.

Q3 How will Ebright's theory of 'how all cells work' benefit human kind?
ans Ebright's theory on how cells can read and use DNA could help scientists understand the causes of certain diseases, develop new treatments, and advance genetic research. His work opened possibilities for preventing genetic disorders and improving human health through deeper knowledge of how life's fundamental instructions operate.

III Answer the following Long Answers:-

Q1 Richard Ebright displayed a well-rounded personality. Do you agree? Ellucidate in the context of the chapter 'Making of a Scientist'.

ans Yes, Richard Ebright displayed a truly well-rounded personality, as shown in 'The Making of a Scientist'. From a young age, he combined curiosity with determination, exploring nature through butterfly collections and scientific experiments. His intellectual hunger went beyond mere hobbies leading him to ask deeper questions and seek meaningful answers.

Ebright was not only academically gifted but also creative and disciplined. He excelled in debating, public speaking, and sports, providing that his talents were not confined to the laboratory. His ability to balance diverse interests reflected his versatility.

Moreover, his humility, perseverance, and willingness to learn from failures made him admirable. These qualities, along with his scientific brilliance, shaped him into a complete individual and an inspiration for aspiring learners.

IV Answer the following word meanings:-

1 sentence Jealous: Showing envy of someone's achievements
She felt jealous of her friend's success.

2 sentences Ornamental: Decorative
Se placed ornamental vases on every shelf.

3 sentences Entomology: Study of insects
He studied entomology to research butterfly species.

4 sentences Sophisticated: complex or refined.
The robot used sophisticated technology to move.

5 sentences Heridity: Traits passed from parents
Eye colour is determined by heredity.